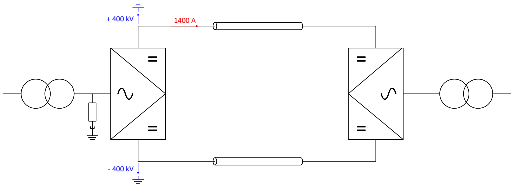
AMERICAN TERAWATT

HVDC Design

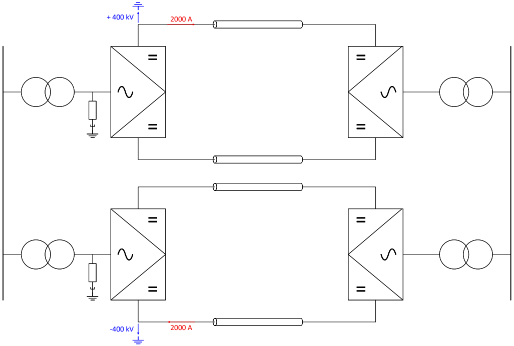
American Terawatt | July 27, 2026

1GW - Symmetrical Monopole (SMP)



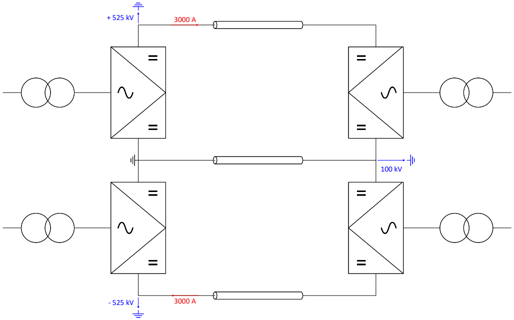
- **Topology** - SMP with $\pm 400kV$
- **DC Line** - Two symmetrical conductors

3GW - Dual Symmetrical Monopole (DSMP)



- **Topology** - DSMP with $\pm 400kV$
- **DC Line** - Four symmetrical conductors

3GW - Bipole (BIP)



- **Topology** - Bipole with $\pm 525kV$
- **DC Line**
  - Two high voltage conductors (HV) and one dedicated metallic return conductor (DMR).
  - In case of cable applications, parallel cables are required due to high current rating.
  - The voltage rating of the DMR depends on the conductor resign and must be refined once the DC line parameters are availabe.
  - For conductor optimization a lower current rating of the DMR can be applied.

Utah/Nevada Industrial Grid (UNIG)

Connection	Name	Topology	Rating
Fennec-Stratos Howell, UT $\rightarrow$ Wells, NV	Tungsten	SMP	1000 MW
Kit-Corsac Hazen, NV $\rightarrow$ Dixie Valley, NV	Copper	DSMP BIP	3000 MW
Marble-Corsac Pronto, NV $\rightarrow$ Hazen, NV	Quartz	SMP	1000 MW
Swift-Blanford Deseret, UT $\rightarrow$ Cherry Creek, NV	Galena	DSMP BIP	3000 MW